

FIBER-GRAND Paper I: Proof Supplement and Claim Boundaries

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Abstract

This supplement gives complete proofs for the narrow Paper-I model: an arbitrary nonempty binary block code, exactly one uniformly located deletion, and independent BSC substitutions. It proves the aggregate likelihood, strict best-history/aggregate-ML reversal, inverse-alignment discovery-shell characterization, exact shell certificate, realization-dependent stopping theorem, finite-length work bounds, and high-probability membership/query exponent. It also records boundary cases and the claims that are intentionally not made.

1 Model and exact likelihood

Let $X = (X_1, \dots, X_n) \in \{0, 1\}^n$. Independent substitutions form $\tilde{X}_i = X_i \oplus N_i$ with $N_i \sim \text{Bernoulli}(p_s)$, $0 \leq p_s < 1/2$. Exactly one coordinate J is deleted uniformly from $\{1, \dots, n\}$. The observation $Y \in \{0, 1\}^m$ has $m = n - 1$. The code $\mathcal{C} \subseteq \{0, 1\}^n$ is arbitrary and nonempty, with uniform prior, and is accessed through a membership test. Define

$$d_j(x, y) = \sum_{i=1}^{j-1} \mathbf{1}\{x_i \neq y_i\} + \sum_{i=j+1}^n \mathbf{1}\{x_i \neq y_{i-1}\}.$$

Proposition 1 (Likelihood and recurrence). *For every x, y ,*

$$W(y \mid x) = \frac{1}{n} \sum_{j=1}^n p_s^{d_j(x, y)} (1 - p_s)^{m - d_j(x, y)}.$$

Moreover,

$$d_1 = \sum_{i=2}^n \mathbf{1}\{x_i \neq y_{i-1}\}, \quad d_{j+1} = d_j - \mathbf{1}\{x_{j+1} \neq y_j\} + \mathbf{1}\{x_j \neq y_j\}.$$

Thus all alignment components are evaluated in $O(n)$ arithmetic operations.

Proof. Condition on $J = j$. The deleted symbol is marginalized, and the m surviving comparisons require exactly d_j substitutions. Independence gives the j th term; averaging over the uniform deletion coordinate gives the likelihood. Moving the hypothesized deletion from j to $j + 1$ changes only the comparison involving y_j : the old suffix term (x_{j+1}, y_j) is removed and the new prefix term (x_j, y_j) is added. This proves the recurrence and complexity statement. $\square$

2 Strict pathwise failure

Let

$$M_y(x) = \max_j \frac{1}{n} p_s^{d_j(x, y)} (1 - p_s)^{m - d_j(x, y)}.$$

Theorem 2 (Strict reversal). *For $n \geq 4$ and $0 < p_s < 1/2$, set*

$$y = 0^{n-3}10, \quad x_A = 0^{n-3}101, \quad x_B = 0^n, \quad \mathcal{C} = \{x_A, x_B\}.$$

Then $M_y(x_A) > M_y(x_B)$, whereas

$$W(y \mid x_B) > W(y \mid x_A) \iff np_s > 1 + 2p_s^2.$$

Proof. Deleting the last coordinate of x_A gives y without substitution, so $M_y(x_A) = n^{-1}(1 - p_s)^m$. Every deletion of x_B leaves 0^m , which differs from y once, so $M_y(x_B) = n^{-1}p_s(1 - p_s)^{m-1}$; the first inequality follows from $p_s < 1/2$. For x_A , mismatch distances 0, 1, 2, 3 occur with multiplicities 1, 1, 1, $n - 3$; every alignment of x_B has distance one. Therefore

$$\begin{aligned} W(y \mid x_B) &= p_s(1 - p_s)^{n-2}, \\ W(y \mid x_A) &= \frac{1}{n}[(1 - p_s)^{n-1} + p_s(1 - p_s)^{n-2} + p_s^2(1 - p_s)^{n-3} + (n - 3)p_s^3(1 - p_s)^{n-4}]. \end{aligned}$$

Direct factorization yields

$$\frac{n(W(y \mid x_B) - W(y \mid x_A))}{(1 - p_s)^{n-4}} = (1 - 2p_s)(np_s - 1 - 2p_s^2).$$

The first factor is positive, proving the equivalence. $\square$

3 Discovery shells

For $z \in \{0, 1\}^m$, let $\mathcal{I}(z)$ be the set of binary length- n words formed by one insertion.

Lemma 3 (One-insertion sphere). *For every z , $|\mathcal{I}(z)| = m + 2 = n + 1$. For inserted bit b , all distinct outputs arise by inserting b in the first gap and immediately after each symbol of z unequal to b .*

Proof. Moving b through a run of b does not change the output. A new output appears initially and each time the insertion crosses a symbol unequal to b , giving $1 + \#\{i : z_i \neq b\}$. Summing over $b = 0, 1$ gives $m + 2$. The two output classes have different Hamming weights and are disjoint. $\square$

Remark 4. Lemma 3 is classical and is used only to remove redundant history-pair generation. It is not asserted as a new coding theorem.

Define the discovery shell $h_y(x) = \min_j d_j(x, y)$.

Proposition 5 (Shell characterization). *When shell s enumerates all error patterns e of weight s , sets $z = y \oplus e$, and generates $\mathcal{I}(z)$, candidate x is generated by the end of shell s iff $h_y(x) \leq s$. Its first shell is $h_y(x)$.*

Proof. If alignment j is hypothesized, deleting x_j gives $z = y \oplus e$ with $\text{wt}(e) = d_j(x, y)$; reinserting x_j at j reconstructs x . Conversely, every generated word comes with such an alignment and error vector. Minimization over j proves both directions. $\square$

4 Certified exact decoder

For $0 < p_s < 1/2$, after completing shell $s < m$, define

$$U_s = p_s^{s+1}(1 - p_s)^{m-s-1}.$$

Theorem 6 (Unseen-candidate certificate). *Every candidate unseen after shell $s < m$ has likelihood at most U_s . If a completely scored incumbent codeword has likelihood strictly greater than U_s , the current incumbent set is the complete global ML tie set. Exhaustion at shell m always returns the exact tie set.*

Proof. By Proposition 5, unseen x satisfies $d_j(x, y) \geq s + 1$ for every j . Since $p_s/(1 - p_s) < 1$, every term in the likelihood is at most $p_s^{s+1}(1 - p_s)^{m-s-1}$; averaging does not change that bound. Strict incumbent comparison excludes both an unseen maximizer and an unseen tie. At shell m , all 2^n candidates have been generated, so no unseen candidate remains. $\square$

At $p_s = 0$, only shell zero has positive probability, and shell zero contains every positive-likelihood candidate. At $p_s = 1/2$, shell ordering provides no strict decay; exhaustive finite termination remains valid, but the strict-gain results below do not apply.

5 Realization-dependent stopping

Let T be the number of substitutions among the m surviving symbols, let $\rho = p_s/(1 - p_s)$, and define

$$L_n(p_s) = \min\{\ell \geq 1 : \rho^\ell < 1/n\} = \left\lfloor \log_{(1-p_s)/p_s} n \right\rfloor + 1.$$

Theorem 7 (Stopping shell). *For $0 < p_s < 1/2$, the exact decoder stops no later than*

$$S \leq \min\{m, T + L_n(p_s) - 1\}.$$

This bound is independent of the code.

Proof. Under the true deletion alignment, the transmitted word has mismatch count T , so it is discovered by shell T and contributes at least $n^{-1}p_s^T(1 - p_s)^{m-T}$. The incumbent is at least this large. At $s = T + L_n - 1$,

$$\frac{U_s}{n^{-1}p_s^T(1 - p_s)^{m-T}} = n\rho^{L_n} < 1,$$

so Theorem 6 certifies by this shell unless finite exhaustion occurs earlier. $\square$

$$\text{Let } \text{Vol}(m, s) = \sum_{w=0}^{\min\{m, s\}} \binom{m}{w}.$$

Corollary 8 (Finite work). *For the realized T , with S as in Theorem 7,*

$$Q_{\text{gen}}, Q_{\text{mem}} \leq (n + 1) \text{Vol}(m, S), \quad Q_{\text{score}} \leq Q_{\text{mem}}.$$

A history-pair generator attempts $2n \text{Vol}(m, S)$ insertion pairs before global deduplication.

Proof. Each error vector contributes exactly $n + 1$ distinct one-insertion supersequences. A global seen set can only reduce membership calls, and only words passing membership receive a complete likelihood evaluation. $\square$

If τ_δ is the smallest binomial quantile satisfying $\mathbb{P}[T \leq \tau_\delta] \geq 1 - \delta$, then with probability at least $1 - \delta$,

$$Q_{\text{mem}} \leq (n + 1) \text{Vol}(m, \min\{m, \tau_\delta + L_n - 1\}).$$

Taking expectation over $T \sim \text{Binomial}(m, p_s)$ gives

$$\mathbb{E}Q_{\text{mem}} \leq (n + 1) \sum_{t=0}^m \binom{m}{t} p_s^t (1 - p_s)^{m-t} \text{Vol}(m, \min\{m, t + L_n - 1\}).$$

6 High-probability query exponent

Let $h_2(u) = -u \log_2 u - (1 - u) \log_2 (1 - u)$.

Theorem 9 (Oracle/query exponent). *Fix $0 < p_s < 1/2$ and $c > 0$. As $n \rightarrow \infty$, with probability at least $1 - n^{-c}$,*

$$Q_{\text{mem}}, Q_{\text{score}} \leq 2^{n(h_2(p_s) + o(1))}.$$

If $|\mathcal{C}_n| = 2^{nR + o(n)}$ with $R > h_2(p_s)$, then exhaustive codeword-wise ML performs at least $2^{n(R - h_2(p_s) - o(1))}$ times as many complete candidate-likelihood evaluations.

Proof. Put $\epsilon_n = \sqrt{c \ln n / (2m)}$. Hoeffding's inequality gives $\mathbb{P}[T > m(p_s + \epsilon_n)] \leq n^{-c}$. Also $L_n = O(\log n)$, so on the complementary event $S/m \leq p_s + o(1) < 1/2$. The standard Hamming-ball bound gives $\text{Vol}(m, S) \leq 2^{mh_2(S/m)} = 2^{n(h_2(p_s) + o(1))}$. The polynomial factor $n + 1$ is absorbed into $2^{o(n)}$, and Corollary 8 proves the first statement. Division of $|\mathcal{C}_n|$ by Q_{score} proves the second. $\square$

7 Claim boundaries

The proofs establish exact aggregate ML, a strict pathwise counterexample, a code-independent stopping shell, and membership/query bounds for the stated one-deletion model. They do not show instance optimality, universal latency superiority, multiple-edit performance, or novelty of path summation, threshold aggregation, GRAND-style membership testing, or insertion-sphere cardinality. The finite campaign contained no naturally sampled case in which the first-shell codeword tie set was disjoint from the aggregate-ML tie set; the strict structural claim rests on Theorem 2, not on empirical prevalence.